

Email Marketing (builder) PRD

Email Builder & Campaign Management

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Overview

The Email Builder enables users to design, send, and analyze marketing email campaigns using a no-code, drag-and-drop interface.

This feature allows users to:

- Create reusable email templates
- Build regular or automated campaigns
- Target audiences using lists and segments
- Schedule email sends
- Allow dynamic variables to be inserted into email templates
- Integrate campaigns into Automations

The Email Builder reuses core components from the Form Builder, including block-based layout configuration, window sizing (**mobile** or **desktop**) and JSON storage structure.

Goals & Objectives

Primary Goals

- Allow users to visually create marketing emails using a block-based builder.
 - Support immediate and scheduled campaign sends.
 - Enable list and segment-based targeting.
 - Integrate campaign sending into the existing Automations execution runtime.
 - Maintain compliance with unsubscribe and suppression rules.
 - **unsubscribed**
 - User opts out
 - **bounced**
 - Email fails **N** number of times
 - **spam_complaint**
 - Recipient marks email as spam
 - **global**
 - Some contacts may be suppressed at the account level (ie. legal disputes, manual suppression, compliance hold, etc.)
-

User Personas

Business Owner / Admin

- Creates brand templates
- Sends newsletters and product updates
- Monitors performance dashboards
- Manages sender domains

Marketing / Sales User

- Creates campaigns
- Builds nurture sequences
- Targets specific segments
- Analyzes open and click performance

Power User / Ops

- Designs complex audience segmentation
 - Integrates campaigns with automations
 - Monitors deliverability and compliance
 - Configures quiet hours and suppression logic
-

Core Concepts

Campaign

A container representing a single email send or automated email sequence.

Attributes:

- name
- status (draft / scheduled / sending / sent / paused)
- type (regular / automated)
- subject line

- preview text
 - sender information
 - audience targeting rules
 - schedule configuration
 - performance metrics
-

Template

Reusable email layout structure.

Attributes:

- name
- category
- tags
- thumbnail
- content_html
- created_by
- timestamps

Templates are block-based and stored as JSON, similar to Form Builder.

List

A saved grouping of contacts.

Attributes:

- id
- name
- subscriber_count
- growth_rate
- avg_open_rate
- avg_click_rate
- tags

Lists do not duplicate contact data. They reference contacts from Contacts Service.

Campaign Execution

Campaign sends are processed by the Automations Execution Runtime.

Each campaign generates execution instances per recipient.

Functional Requirements

Campaign Lifecycle

Users can:

- Create a new campaign
- Rename a campaign
- Save as draft

- Schedule campaign
- Send immediately
- Pause campaign
- Duplicate campaign
- Delete campaign (with confirmation)

Only scheduled or active campaigns are processed for sending.

Email Builder UI

Canvas

- Block-based drag-and-drop builder
- Reuses Form Builder components
- Supports:
 - Heading blocks
 - Text blocks
 - Image blocks
 - Video blocks
 - Button blocks
 - Columns blocks
 - Divider blocks
 - Spacer blocks
 - Html blocks
 - Discount Code blocks
 - Menu blocks
 - Social Links blocks
- Mobile preview mode
- Desktop preview mode

Properties Panel

Each block supports configuration:

- Styling (font, color, padding)
 - Links
 - Personalization tokens (e.g., {{first_name}})
 - Visibility conditions
 - Only show specific sections for certain **conditions** or **list** (ie. **VIP Sales List** or **leadScore < 40**)
-

Campaign Configuration

Sender Configuration

- sender_name
 - sender_email
 - reply_to
 - Domain must be verified
 - DKIM/SPF enforcement
-

Subject & Preview

- Subject line required
 - Preview text optional
 - Character count indicators
 - Rich-text support (**bold**, **font-color**, **alignment**, etc.)
-

Audience Targeting

Users can target campaigns by:

- List selection (multiple allowed)
- Segment selection
- Excluding unsubscribed contacts
- Respecting suppression list

Audience preview must show:

- Estimated total recipients
-

Scheduling

Campaign supports:

- Send immediately
- Schedule at specific datetime
- Timezone selection
- Respect quiet hours (per contact)
 - Expect a **contact** json will have a **quietHoursStart** and **quietHoursEnd** field with **ISO8601** using UTC timezone (eg. `2025-04-15T14:30:00Z`)

Quiet hours logic:

- If enabled, execution delayed until allowed time window
-

Compliance

System must:

- Automatically inject unsubscribe link
 - Maintain suppression list
 - Respect do-not-contact flags
 - Prevent sending to unsubscribed contacts
 - Log unsubscribe events
-

Automations Integration (back-end)

Campaigns can be:

- Sent manually
- Triggered via Automation
- Used as an action node inside Automation Builder

When triggered by automation:

- Campaign execution runs inside Execution Runtime

- Execution ends at campaign completion
 - Performance metrics aggregate normally
-

Data Model

Campaign Object

- id
 - name
 - status
 - type
 - subject_line
 - preview_text
 - sender_name
 - sender_email
 - reply_to
 - audience
 - schedule
 - performance
 - created_at
 - updated_at
-

Template Object

- id
 - name
 - category
 - tags[]
 - thumbnail_url
 - content_html
 - created_by
 - created_at
 - updated_at
-

List Object

- id
 - name
 - subscriber_count
 - growth_rate_30d
 - avg_open_rate
 - avg_click_rate
 - tags[]
 - created_at
-

Campaign Recipient Log

- campaign_id

- contact_id
- status
- sent_at
- delivered_at
- opened_at
- clicked_at
- bounced
- unsubscribed

Sample Payload

```
1  {
2    "campaigns": [
3      {
4        "id": "cmp_01",
5        "name": "Welcome Series - New Subscribers",
6        "status": "active",
7        "type": "automated",
8        "subject_line": "Welcome to the family! Here is your 20% off
code.",
9        "preview_text": "Get started with your exclusive discount.",
10       "sender_name": "Sarah from Acme Corp",
11       "sender_email": "sarah@acmecorp.com",
12       "reply_to": "support@acmecorp.com",
13       "audience": {
14         "list_ids": ["lst_01"],
15         "segment_ids": ["seg_new_this_week"],
16         "total_recipients": 1250
17       },
18       "schedule": {
19         "send_at": "immediately",
20         "timezone": "UTC",
21         "respect_quiet_hours": true
22       },
23       "performance": {
24         "sent": 1250,
25         "delivered": 1245,
26         "bounced": 5,
27         "opens": 850,
28         "open_rate": 68.2,
29         "clicks": 420,
30         "click_rate": 33.6,
31         "unsubscribes": 2,
32         "spam_reports": 0
33       },
34       "created_at": "2024-01-15T08:00:00Z",
35       "updated_at": "2024-01-15T08:00:00Z"
36     },
37     {
38       "id": "cmp_02",
39       "name": "Q3 Product Update Newsletter",
40       "status": "draft",
41       "type": "regular",
42       "subject_line": "New Features: AI Automation & Advanced
Reporting",
43       "preview_text": "See what we've been building for you.",
44       "sender_name": "Acme Corp Product Team",
45       "sender_email": "hello@acmecorp.com",
46       "reply_to": "hello@acmecorp.com",
47       "audience": {
48         "list_ids": ["lst_01", "lst_02"],
49         "segment_ids": [],
50         "total_recipients": 15000
51       },
52       "schedule": {
53         "send_at": "2024-03-01T10:00:00Z",
54         "timezone": "America/New_York",
55         "respect_quiet_hours": false
56       },
57       "performance": {
58         "sent": 0,
59         "delivered": 0,
60         "bounced": 0,
61         "opens": 0,
62         "open_rate": 0,
63         "clicks": 0,
64         "click_rate": 0,
65         "unsubscribes": 0,
```

```
66         "spam_reports": 0
67     },
68     "created_at": "2024-02-20T14:30:00Z",
69     "updated_at": "2024-02-22T09:15:00Z"
70 }
71 ],
72 "templates": [
73     {
74         "id": "tpl_01",
75         "name": "Standard Newsletter V2",
76         "category": "newsletter",
77         "tags": ["monthly", "update", "company"],
78         "thumbnail_url": "https://placeholder.co/300x400/png",
79         "content_html": "<!DOCTYPE html><html><body>...</body></html>",
80         "created_by": "usr_789",
81         "created_at": "2023-11-10T11:20:00Z",
82         "updated_at": "2024-01-05T16:45:00Z"
83     },
84     {
85         "id": "tpl_02",
86         "name": "E-commerce Abandoned Cart",
87         "category": "transactional",
88         "tags": ["ecommerce", "abandoned_cart", "recovery"],
89         "thumbnail_url": "https://placeholder.co/300x400/png",
90         "content_html": "<!DOCTYPE html><html><body>...</body></html>",
91         "created_by": "usr_123",
92         "created_at": "2023-09-05T09:00:00Z",
93         "updated_at": "2023-09-05T09:00:00Z"
94     }
95 ],
96 "lists": [
97     {
98         "id": "lst_01",
99         "name": "Master Subscriber List",
100         "subscriber_count": 45200,
101         "growth_rate_30d": 5.2,
102         "avg_open_rate": 24.5,
103         "avg_click_rate": 4.1,
104         "tags": ["primary", "all-users"],
105         "created_at": "2022-01-01T00:00:00Z"
106     },
107     {
108         "id": "lst_02",
109         "name": "VIP Customers (LTV > $500)",
110         "subscriber_count": 3450,
111         "growth_rate_30d": 1.5,
112         "avg_open_rate": 58.7,
113         "avg_click_rate": 18.2,
114         "tags": ["vip", "high-value"],
115         "created_at": "2022-06-15T00:00:00Z"
116     }
117 ],
118 "contacts": [
119     {
120         "id": "cnt_1001",
121         "email": "alex.johnson@example.com",
122         "first_name": "Alex",
123         "last_name": "Johnson",
124         "status": "subscribed",
125         "rating": 4.5,
126         "lists": ["lst_01", "lst_02"],
127         "tags": ["webinar-attendee", "purchased-q1"],
128         "custom_fields": {
129             "company": "TechSolutions Inc",
130             "job_title": "Marketing Director",
131             "lifetime_value": 850.00
132         },
133         "engagement": {
134             "total_sent": 45,
135             "total_opens": 38,
136             "total_clicks": 12,
137             "last_opened_at": "2024-02-20T10:15:00Z",
138             "last_clicked_at": "2024-02-15T14:22:00Z"
139         },
140         "created_at": "2023-03-10T08:30:00Z",
141         "updated_at": "2024-02-20T10:15:00Z"
142     },
143     {
144         "id": "cnt_1002",
145         "email": "m.smith@startup.io",
146         "first_name": "Maria",
147         "last_name": "Smith",
148         "status": "unsubscribed",
149         "rating": 1.0,
150         "lists": ["lst_01"],
151         "tags": ["lead-gen-2023"],
152         "custom_fields": {
153             "company": "Startup.io",
```



```
154         "job_title": "Founder",
155         "lifetime_value": 0.00
156     },
157     "engagement": {
158         "total_sent": 12,
159         "total_opens": 2,
160         "total_clicks": 0,
161         "last_opened_at": "2023-11-05T09:00:00Z",
162         "last_clicked_at": null
163     },
164     "created_at": "2023-10-01T11:00:00Z",
165     "updated_at": "2023-12-15T16:00:00Z"
166 }
167 ],
168 "automations": [
169     {
170         "id": "aut_01",
171         "name": "Abandoned Cart Recovery",
172         "status": "active",
173         "trigger": "cart_abandoned",
174         "steps_count": 5,
175         "active_contacts": 142,
176         "performance": {
177             "completed": 850,
178             "revenue_recovered": 12500.50,
179             "conversion_rate": 18.5
180         },
181         "created_at": "2023-08-20T10:00:00Z"
182     }
183 ]
184 }
```

Execution Model (High-Level)

1. Campaign created
2. Audience resolved
3. Execution records created per recipient
4. Execution Runtime processes recipients in batches
5. Vendor API sends email
6. Webhooks update performance data
7. Campaign status updated when complete

Each execution maintains:

- contact reference
- campaign reference
- delivery state
- timestamps

Error Handling & Safety

- Failed sends logged with error reason
- Bounced emails marked for suppression
- Campaign can be paused mid-send
- Duplicate send prevention enforced
- Vendor rate limits respected

Permissions

- Only users with **EMAIL_MARKETING_EDIT** permissions can create/edit campaigns
 - View-only **EMAIL_MARKETING** users can view performance but not modify campaigns
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Success Metrics

- % of active accounts sending at least one campaign
- Average campaigns per account
- Open rate benchmark improvement
- Reduction in manual follow-up emails
- Delivery success rate > 98%